EXISTENCE OF THE DERIVED FUNCTOR L-ETA_I COMPUTED BY THE ETA CONSTRUCTION ON COMPLEXES OF I-TORSION FREE MODULES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span contains no commutative diagram and cites only cross-chapter results, each recorded in assembly.external_references with its canonical tag; no stub replaces a load-bearing argument.

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Allowed standard prerequisites (editorial).

- Localization of a full triangulated subcategory at quasi-isomorphisms. Stacks Project, Derived Categories, Lemma derived-lemma-localization-subcategory (tag 0GSL). The criterion that reduces the equivalence S^{-1}K(T) = D(O_X) to the existence of torsion-free resolutions.
- Existence of K-flat resolutions. Stacks Project, Cohomology of Sheaves, Lemma lemma-K-flat-resolution (tag 06YF). Provides the flat (hence I-torsion free) complex quasi-isomorphic to a given complex.
- Characterizations of I-torsion freeness and quasi-isomorphism invariance of eta. Stacks Project, Cohomology of Sheaves, Lemmas lemma-I-torsion-free (tag 0GT5) and lemma-eta-qis (tag 0F8P). The third characterization links flatness to torsion freeness, and eta-qis is what makes the functor descend to the derived category.

Source reference. Original reference 0GT4: sources/stacks-chow/tags.txt, source lines 20954–20954. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GSL: sources/stacks-chow/tags.txt, source lines 20936–20936. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06YF: sources/stacks-chow/tags.txt, source lines 8884–8884. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GT5: sources/stacks-chow/tags.txt, source lines 20955–20955. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0F8P: sources/stacks-chow/tags.txt, source lines 19049–19049. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 04VG: sources/stacks-chow/tags.txt, source lines 6324–6324. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0GT4 there is an additive functor*[1] $L\eta_{\mathcal{I}} : D(\mathcal{O}_X) \to D(\mathcal{O}_X)$ *such that if* M *in* $D(\mathcal{O}_X)$ *is represented by a complex* $\mathcal{F}^\bullet$ *of* $\mathcal{I}$*-torsion free* $\mathcal{O}_X$*-modules, then* $L\eta_{\mathcal{I}} M = \eta_{\mathcal{I}} \mathcal{F}^\bullet$*. Similarly for morphisms.*

Proof. Denote $\mathcal{T} \subset Mod(\mathcal{O}_X)$ the full subcategory of $\mathcal{I}$-torsion free $\mathcal{O}_X$-modules. We have a corresponding inclusion

$$K(\mathcal{T}) \quad \subset \quad K(Mod(\mathcal{O}_X)) = K(\mathcal{O}_X)$$

of $K(\mathcal{T})$ as a full triangulated subcategory of $K(\mathcal{O}_X)$. Let $S \subset \text{Arrows}(K(\mathcal{T}))$ be the quasi-isomorphisms. We will apply Derived Categories, Lemma 0GSL to show that the map

$$S^{-1}K(\mathcal{T}) \longrightarrow D(\mathcal{O}_X)$$

is an equivalence of triangulated categories. The lemma shows that it suffices to prove: given a complex $\mathcal{G}^\bullet$ of $\mathcal{O}_X$-modules, there exists a quasi-isomorphism $\mathcal{F}^\bullet \to \mathcal{G}^\bullet$ with $\mathcal{F}^\bullet$ a complex of $\mathcal{I}$-torsion free $\mathcal{O}_X$-modules. By Lemma 06YF we can find a quasi-isomorphism $\mathcal{F}^\bullet \to \mathcal{G}^\bullet$ such that the complex $\mathcal{F}^\bullet$ is K-flat (we won't use this) and consists of flat $\mathcal{O}_X$-modules $\mathcal{F}^i$. By the third characterization of Lemma 0GT5 we see that a flat $\mathcal{O}_X$-module is an $\mathcal{I}$-torsion free $\mathcal{O}_X$-module and we are done.

With these preliminaries out of the way we can define $L\eta_f$. Namely, by the discussion following Lemma 0GT5 this section we have already a well defined functor

$$K(\mathcal{T}) \xrightarrow{\eta_f} K(\mathcal{T}) \to K(\mathcal{O}_X) \to D(\mathcal{O}_X)$$

which according to Lemma 0F8P sends quasi-isomorphisms to quasi-isomorphisms. Hence this functor factors over $S^{-1}K(\mathcal{T}) = D(\mathcal{O}_X)$ by Categories, Lemma 04VG. □

[1] Beware that this functor isn't exact, i.e., does not transform distinguished triangles into distinguished triangles.